

SIMATS ENGINEERING

ASSESSMENT TOOL 2

COURSE NAME: Artificial Intelligence

COURSE CODE: CSA17

COURSE OUTCOME COVERED

CO2: Experiment search and game-solving techniques such as Greedy Search, A*, CSP, Minimax, and Alpha-Beta pruning with heuristic functions for solving complex AI problems efficiently. (BL3)

Assessment Tool Weightage: 50%

QUESTIONS: 5

Duration: 1 Hour
Total Marks:50

Annexure A

Scenario Based Assignment

Code of Conduct:

I mohammed arsath(192511130) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

Mohammed arsath m

Annexure A

Scenario Based Assignment

1. **Scenario** : A government deploys AI-powered drones to deliver emergency medicines in a flood-affected region. The environment is highly dynamic—roads may suddenly become inaccessible, and weather conditions affect travel cost. The drone has access to: A heuristic estimate (straight-line distance), Partial real-time updates about blocked paths and Variable movement costs depending on terrain and weather. However, due to urgency, the drone must quickly decide routes with minimum delay and risk.

Tasks:

- i. Explain how Greedy Best-First Search would behave in this scenario and discuss its strengths and weaknesses under uncertainty
- ii. Explain how A* would handle the same situation, including how it balances cost and heuristic
- iii. Analyze the impact of heuristic accuracy on both algorithms
- iv. Discuss how dynamic changes (blocked paths) affect search performance
- v. Recommend the most suitable algorithm and justify your decision considering real-time constraints, optimality, and safety

Answer:

Greedy Best-First Search (GBFS) selects the path that appears closest to the destination by using only the heuristic value, such as the straight-line distance. In this emergency medicine delivery scenario, the drone will always choose the route that seems nearest to the target without considering the actual travel cost or possible risks. This allows the drone to make decisions quickly, which is useful during emergencies.

Strengths:

Makes fast routing decisions.

Requires less computation and memory.

Suitable for time-critical situations.

Weaknesses:

Ignores actual travel cost and obstacles.

May choose blocked or unsafe paths.

Does not guarantee the shortest or safest route.

Performance decreases when heuristic information is inaccurate.

- ii). A* Search uses both the actual cost from the starting point $g(n)$ and the estimated cost to the destination $h(n)$. It evaluates each possible path using:

$$f(n) = g(n) + h(n)$$

In this scenario, A* considers weather conditions, terrain difficulty, and blocked roads while also estimating the remaining distance. This enables the drone to select a route that minimizes both travel time and risk. If new information about blocked paths becomes available, A* can recalculate and choose a better alternative path.

- iii. Analyze the impact of heuristic accuracy on both algorithms

The quality of the heuristic greatly affects both search algorithms.

If the heuristic is highly accurate, Greedy Search reaches the destination quickly with fewer explored nodes, while A* finds the optimal path efficiently.

If the heuristic is inaccurate, Greedy Search may follow poor or blocked routes because it depends only on the heuristic value. A* is less affected because it also considers the actual path cost, making it more reliable and accurate.

- iv. Discuss how dynamic changes (blocked paths) affect search performance

In a flood-affected region, roads may suddenly become blocked or unsafe.

Greedy Best-First Search may repeatedly choose blocked routes because it only focuses on the estimated distance. This increases replanning time and may delay medicine delivery.

A* can update the search when new information is received, recalculate the path cost, and identify a safer alternative route. Although this requires more computation, it improves reliability and safety.

- v. Recommend the most suitable algorithm and justify your decision considering real-time constraints, optimality, and safety

A Search* is the most suitable algorithm for this scenario.

Justification:

Produces an optimal route by considering both actual cost and estimated distance.

Adapts better to dynamic changes such as blocked roads and changing weather.

Provides safer navigation compared to Greedy Search.

Balances speed, optimality, and reliability, making it suitable for real-time emergency medicine delivery in uncertain environments.

2. Scenario: A smart city implements an AI system to optimize traffic signal timings across hundreds of intersections. The objective is to minimize Total waiting time, Fuel consumption and Traffic congestion. The system must continuously adjust signals based on

traffic flow. However the search space is extremely large, traffic patterns change dynamically and the system may get stuck in locally optimal solutions

Tasks. Formulate this as an optimization problem and explain why traditional search is inefficient

- i. Explain how Hill Climbing would work in this scenario and identify its limitations
- ii. Discuss issues like local maxima, plateaus, and ridges with real-world interpretation
- iii. Explain how Simulated Annealing or Genetic Algorithms can overcome these limitations
- iv. Recommend the best approach and justify your choice based on scalability and adaptability

Answer:

- i. Formulate this as an optimization problem and explain why traditional search is inefficient

The traffic signal optimization problem aims to minimize waiting time, fuel consumption, and traffic congestion by finding the best signal timings for all intersections. The AI system must also adapt to continuously changing traffic conditions.

Traditional search methods are inefficient because the number of possible signal timing combinations is extremely large. As the number of intersections increases, the search space grows rapidly, making it difficult to find the best solution within a short time.

- ii. Explain how Hill Climbing would work and identify its limitations

Hill Climbing starts with an initial traffic signal plan and makes small changes to improve traffic flow. It selects the neighboring solution that gives the best improvement and continues until no better solution is found.

Limitations:

May get stuck in a local optimum.

Cannot guarantee the global best solution.

Performance depends on the starting point.

Less effective when traffic conditions change frequently.

- iii. Discuss local maxima, plateaus, and ridges

Local Maxima: The algorithm finds a good solution but not the best one. For example, one intersection improves while nearby roads remain congested.

Plateaus: Several solutions have the same performance, making it difficult for the algorithm to decide the next move.

Ridges: Better solutions require multiple changes together, but Hill Climbing changes only one step at a time.

- iv. Explain how Simulated Annealing or Genetic Algorithms overcome these limitations

Simulated Annealing accepts some worse solutions during the search, helping it escape local maxima and explore better paths.

Genetic Algorithm uses selection, crossover, and mutation to generate better traffic signal plans. It explores many solutions simultaneously and adapts well to changing traffic conditions.

- v. Recommend the best approach

The Genetic Algorithm is the most suitable approach because it efficiently handles large search spaces, avoids local optima, and adapts to dynamic traffic patterns. It provides better optimization for real-time smart city traffic management while reducing congestion, waiting time, and fuel consumption.

3. Scenario: An autonomous rover is deployed on Mars to explore unknown terrain. It must: navigate safely to collect samples, avoid hazards (craters, rocks), operate with limited sensing capability and make decisions without a complete map. The rover updates its knowledge as it explores, making this a real-time decision-making problem.

Tasks:

- i. Explain why this problem requires an online search agent instead of offline search
- ii. Analyze the challenges of partial observability and dynamic environments
- iii. Describe how the rover builds and updates its internal model
- iv. Compare online search with uninformed and informed search strategies
- vi. Justify the most suitable approach considering uncertainty, adaptability, and safety

Answer:

- i. Explain why this problem requires an online search agent instead of offline search

The Mars rover explores an unknown environment where a complete map is not available. It must make decisions while moving and update its path based on new information. Therefore, an online search agent is more suitable because it learns and acts at the same time.

- ii. Analyze the challenges of partial observability and dynamic environments

The rover has limited sensors and cannot detect all obstacles in advance. It may encounter rocks, craters, or rough terrain unexpectedly. Since the environment is uncertain, the rover must continuously adjust its path to avoid hazards and safely reach its destination.

- iii. Describe how the rover builds and updates its internal model

The rover collects information through cameras and sensors while exploring. It stores the discovered paths and obstacles in its internal map. As new information becomes available, it updates the map and chooses a safer and more efficient route.

iv. Compare online search with uninformed and informed search strategies

Online Search: Explores and makes decisions in real time using newly discovered information.

Uninformed Search: Searches without heuristic knowledge and explores many unnecessary paths.

Informed Search: Uses heuristic information to find better paths but usually assumes the environment is already known.

v. Justify the most suitable approach considering uncertainty, adaptability, and safety

Online Search is the best approach because it can adapt to unknown environments, respond to unexpected obstacles, and update its decisions continuously. This improves the rover's safety, flexibility, and efficiency during Mars exploration

4. Scenario: A university is designing an AI system to automatically generate exam timetables. The system must satisfy multiple constraints: No student should have overlapping exams, Limited number of exam halls, Certain subjects must be scheduled before others, Faculty availability constraints, Special accommodations for some students, The complexity increases as the number of courses and students grows.

Tasks :

Formulate the problem as a Constraint Satisfaction Problem (CSP)

- i. Clearly define variables, domains, and constraints with explanation
- ii. Explain how backtracking search works in this scenario
- iii. Discuss how constraint propagation (e.g., forward checking) improves efficiency
- iv. Analyze real-world challenges and justify the best strategy for scalability

Answer:

- i. Define variables, domains, and constraints

The exam timetable can be modeled as a Constraint Satisfaction Problem (CSP).

Variables: Exams for each subject.

Domains: Available dates, time slots, and exam halls.

Constraints: No student should have overlapping exams, exam hall capacity should not be exceeded, faculty must be available, prerequisite subjects should be scheduled first, and special accommodations must be provided for eligible students.

- ii. Explain how backtracking search works

Backtracking assigns a time slot and hall to one exam at a time. If a constraint is violated, such as overlapping exams or unavailable faculty, the algorithm goes back to the previous assignment and tries another option. This process continues until a valid timetable is generated.

- iii. Explain how constraint propagation (Forward Checking) improves efficiency

Forward checking removes invalid choices from the remaining variables after each assignment. This reduces the search space and detects conflicts early, allowing the algorithm to generate a timetable faster with fewer backtracking steps.

- iv. Analyze real-world challenges and justify the best strategy

In real-world universities, the number of courses, students, and constraints increases every semester, making timetable generation more complex. Combining Backtracking with Forward Checking is the best strategy because it reduces unnecessary searches, satisfies all constraints efficiently, and produces a valid timetable within less time.

5. Scenario: Strategic Game AI for Real-Time Decision Making (Minimax & Alpha-Beta Pruning) A gaming company is developing an AI opponent for a real-time strategy game. The AI must complete against human players, Make decisions within strict time limits, Evaluate complex game states with multiple possible moves, The decision tree is extremely large, making computation expensive.

Tasks:

- i. Explain how the Minimax algorithm models this problem
- ii. Analyze the computational challenges of large game trees
- iii. Explain how Alpha-Beta pruning improves efficiency with detailed reasoning
- iv. Design an evaluation function and justify the factors considered
- v. Recommend a strategy for balancing optimality and real-time performance

Answer:

- i. Explain how the Minimax algorithm models this problem

The Minimax algorithm models the game as a decision tree where the AI tries to maximize its score while assuming the human player makes the best possible moves to minimize it. The AI evaluates all possible moves and selects the one that provides the highest advantage.

- ii. Analyze the computational challenges of large game trees

In real-time strategy games, each move creates many possible future moves, resulting in a very large game tree. Exploring every possible state requires high computation and memory, making it difficult to make decisions within the available time.

iii. Explain how Alpha-Beta Pruning improves efficiency

Alpha-Beta Pruning improves Minimax by eliminating branches that cannot influence the final decision. This reduces the number of nodes evaluated, speeds up the search process, and allows the AI to make faster decisions without affecting the final result.

iv. Design an evaluation function and justify the factors considered

The evaluation function can consider factors such as player score, available resources, army strength, territory control, and remaining health. These factors help the AI estimate the strength of each game state and choose the best move.

v. Recommend a strategy for balancing optimality and real-time performance

The best approach is to use Minimax with Alpha-Beta Pruning. It provides strong decision-making while reducing unnecessary computations, allowing the AI to respond quickly and effectively in real-time strategy games.

RUBRICS FOR CALCULATION:

Criteria	Weightage (%)	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Understanding of Scenario	20%	Demonstrates clear and complete understanding of the scenario and context	Shows good understanding with minor gaps	Partial understanding; misses key aspects	Misinterprets or fails to understand the scenario
Identification of Key Issues	20%	Accurately identifies all relevant issues and factors involved	Identifies most key issues with minor omissions	Identifies limited issues; some irrelevant points	Unable to identify key issues
Application of Concepts	25%	Applies appropriate concepts effectively to address the scenario	Applies concepts with minor errors	Limited or partially correct application	Unable to apply relevant concepts
Decision Making	20%	Makes well-reasoned, logical decisions with strong rationale	Makes reasonable decisions with some justification	Makes weak decisions with limited justification	No clear decisions made or rationale provided
Clarity of Response	15%	Response is clear, structured, and logically presented	Generally clear with minor issues	Somewhat unclear or poorly structured	Disorganized and difficult to understand